

Modernizing Kenya's Judicial System Through Data Intelligence

Across Kenya's courtrooms, a familiar scene plays out each day. Judges like Justice Achieng spend long hours sifting through old files, piecing together facts from documents stored in different systems or still held in physical form. Many of these cases were filed years earlier, yet they continue to consume valuable time because the information needed to resolve them is scattered or incomplete. This is not an isolated experience—it mirrors a judicial system straining under the weight of inefficiency. By 2023, the country had accumulated more than 500,000 pending cases, with most disputes taking three to five years to conclude. Certain land and commercial matters stretch far longer, leaving citizens and businesses in limbo.

The impact of these delays reaches beyond the courts. Kenya ranked 164th globally for contract enforcement in the World Bank Doing Business 2020 report, which estimates that settling a commercial dispute requires 465 days and 44 procedural steps. A study by the University of Nairobi's School of Law estimates that slow case resolution drains approximately KES 2.3 billion every month from the economy due to frozen investments, unresolved transactions, and lost opportunities.

Much of this crisis comes down to how judicial data is created, handled, and used. The challenges can be understood through four key stages:

1. Data Creation & Capture – Slow and Error-Prone Digitization

A significant portion of court records—ranging from pleadings and evidence bundles to judgments—still exists on paper or in partially digitized formats. Court staff spend large amounts of time manually entering information, which leads to 5–10% error rates and frequent misfiling. Handwritten notes, low-quality scans, and bilingual documentation (English and Kiswahili) introduce further complications.

2. Data Validation & Trust – Difficulty Verifying Authentic Documents

Forgery and tampered documents remain a major issue, contributing to nearly one in five case dismissals. Confirming the legitimacy of affidavits, certificates, and other filings typically requires reaching out to agencies such as KRA, NTSA, or county land registries—a process that can take days. Without tools for quick verification, courts face delays that open the door to manipulation and undermine confidence in outcomes.

3. Data Access & Utilization – Information Locked in Silos

Judicial data is spread across multiple platforms that do not communicate with one another. This makes it difficult for judges and lawyers to locate case histories, past rulings, or relevant statutes. The result is inconsistent decisions, repeated hearings, and administrative bottlenecks that could be avoided with unified access to information.

4. Data Intelligence – Limited Use of Analytics and Forecasting

Kenya's judiciary currently lacks the analytical tools needed to forecast case timelines, assess resource needs, or identify emerging patterns. Without this visibility, the system remains reactive. Workloads vary dramatically across courts, timelines are unpredictable, and planning for future demand becomes guesswork.

This hackathon project aims to create practical, AI-enabled tools that simplify digitization, strengthen document verification, integrate judicial data, and provide meaningful predictive insights—helping Kenya's courts deliver faster, fairer, and more transparent justice.

Solution: Sheria — An AI-Driven Judicial Data Intelligence Platform

Sheria is a modern governance platform designed to help Kenya's judiciary tackle delays, improve accuracy, and strengthen trust in legal processes. It brings together digitization, verification, search, and predictive analytics into a single, unified system that fits naturally into everyday judicial workflows. The goal is simple: reduce the time spent looking for information and increase the time spent delivering justice.

Sheria addresses the four core challenges in the justice system—digitization bottlenecks, document authenticity, fragmented data, and lack of analytics—through the following integrated modules:

1. Digitize Documents — Automated OCR, Classification & Metadata Extraction

Courts still rely heavily on handwritten notes, scanned pages, and paper files. Sheria turns these into clean, searchable digital records using:

Advanced OCR trained on Kenyan handwriting, stamps, and legal templates

Automatic classification of documents (affidavits, witness statements, rulings, evidence bundles, etc.)

Entity extraction for names, dates, land parcel numbers, case numbers, and legal citations

Multi-language support (English, Kiswahili, Sheng variants)

Judicial staff can digitize an entire file in minutes, reducing manual data entry and eliminating common errors like misfiling or missing pages. Once digitized, documents become instantly searchable across the entire platform.

2. Verify & Validate — Real-Time Document Authentication

Sheria introduces a fast and reliable way to confirm the legitimacy of affidavits, certificates, land documents, and other records submitted in court. It uses:

Integration cross-checks with trusted registries (KRA, NTSA, NLC, eCitizen, company registries)

AI-driven fraud detection to spot anomalies, duplicates, tampered pages, or digital edits

A backend audit trail that preserves document integrity from submission to judgment

This eliminates days of back-and-forth with external agencies, reduces fraudulent filings, and helps protect the integrity of proceedings.

3. Ask — Conversational Search Across Laws, Cases & Court Records

Instead of manually digging through multiple databases, judges and lawyers can ask Sheria questions in natural language, such as:

“Show me all land disputes involving adverse possession between 2018 and 2023.”

“What precedent exists for cancellation of forged title deeds?”

“Summaries of related cases with similar facts to this one.”

The system retrieves documents, highlights cited sources, and provides context—helping judges make consistent, well-supported decisions.

4. Predict — Analytics & Forecasting for Judicial Efficiency

Sheria’s predictive engine gives court leaders the ability to plan and act proactively. It provides:

Forecasted timelines for each case based on complexity, history, and court workload

Models estimating the risk of case stalling or being dismissed

Resource optimization tools showing where courts are overloaded or underutilized

Trend analysis for emerging legal disputes (e.g., land fraud hotspots, commercial defaults)

This supports evidence-based planning and reduces unpredictable delays.

Impact

By integrating these capabilities, Sheria can help Kenya's judiciary:

Cut document processing times from hours to seconds

Reduce verification delays from days to less than a minute

Improve accuracy and consistency of judgments

Accelerate clearance of the backlog

Strengthen public trust through transparency and efficiency

Sheria is not a replacement for judicial experience—it is a tool that frees judges, registrars, and lawyers from repetitive work so they can focus on what matters most: fair, timely justice.

Technology Stack

Client Side(Frontend): Next.js, Typescript

Server Side(BackEnd): Python, FastAPI

Database: ChromaDb(Vector), PostgreSQL(RDMS)

Artificial Intelligence/Machine Learning: Qwen LLM, Scikit-Learn

Data Capture: Tesseract OCR

Reverse Proxy: Nginx

Orchestration: Docker

Storage: MinIO

Task Processing: Redis, Celery

Open Source ML: LangChain, Ollama

Relevance to Theme: AI for National Prosperity

Kenya's progress as a modern, competitive nation depends on a justice system that is efficient, trusted, and accessible. Today, however, more than half a million unresolved court cases, slow verification processes, and inconsistent access to legal information continue to strain institutions, delay development projects, and weaken public confidence. The Sheria Platform directly targets these challenges by applying artificial intelligence to strengthen governance—one of the core pillars of national prosperity.

At its heart, the solution enhances the judiciary's capacity to process information, verify documents, and make timely, evidence-driven decisions. By automating the digitization of case files using OCR, streamlining document verification through AI-based fraud detection, and

enabling judges to search and analyze legal materials through a smart knowledge engine, Sheria reduces delays that have traditionally cost the country billions of shillings in stalled investments and prolonged disputes. Faster justice directly supports economic growth by unlocking land transactions, accelerating commercial resolutions, and reducing uncertainty for businesses operating in Kenya.

The platform also strengthens public sector governance—a key national priority under the Digital Transformation Agenda. Sheria introduces transparent audit trails, reliable document authentication, and consistent access to judicial data across courts. This reduces opportunities for manipulation, improves accountability, and ensures that decisions are based on verified records rather than missing files or incomplete information. By lowering the error rates associated with manual data entry and fragmented systems, the solution improves both the quality and fairness of judicial outcomes.

In terms of sustainability, Sheria promotes long-term institutional resilience. Digitizing court archives reduces the reliance on paper files, minimizes physical storage costs, and preserves records in formats that can be accessed for generations. Predictive analytics further enable court administrators to forecast case volumes, allocate resources more effectively, and reduce bottlenecks before they occur. This moves the justice system from a reactive model to a forward-looking, data-driven institution—one that can scale with population growth and rising demands on public services.

The platform also contributes to national security by safeguarding the integrity of legal documents. Fraudulent affidavits, forged land records, and tampered evidence are major threats to judicial credibility and have, in many cases, fueled land conflicts and economic losses. Sheria's real-time verification capabilities, powered by AI and blockchain-based audit trails, significantly weaken these risks. A more secure justice system promotes social stability and decreases the likelihood of disputes escalating into broader community or economic conflicts.

Overall, the Sheria Platform embodies the theme **“AI for National Prosperity”** by addressing a systemic challenge that affects every sector—from land and housing to business, investment, and public trust. By combining AI, data intelligence, and digitization, the solution strengthens governance, accelerates economic activity, enhances national security, and lays the foundation for a more just, efficient, and prosperous Kenya.

Presentation: <https://presentationhackon2025.netlify.app/>